

Daily Practice AP Calculus Unit 4

Topics: Contextual Applications of Differentiation (Related Rates, PVA, Linear Approximation, L'Hôpital's Rule)

Section 1: Multiple Choice Questions

No calculator is allowed for these questions.

1. What is the value of $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\ln(x-1)}$?

- (A) 0
- (B) 2
- (C) 4
- (D) The limit does not exist.

2. A particle moves along the x-axis such that its position at time $t \geq 0$ is given by $x(t) = \frac{1}{3}t^3 - 3t^2 + 8t$. On which of the following intervals is the particle moving to the left?

- (A) $0 < t < 2$
- (B) $2 < t < 4$
- (C) $t > 4$
- (D) $0 < t < 2$ and $t > 4$

3. Let f be a twice-differentiable function such that $f(2) = 5$ and $f'(2) = 3$. It is known that $f''(x) < 0$ for all x . Using the tangent line to the graph of f at $x = 2$, which of the following is the best approximation for $f(2.1)$ and describes its relationship to the true value?

- (A) 5.3, which is an underestimate.
- (B) 5.3, which is an overestimate.
- (C) 5.03, which is an underestimate.
- (D) 5.03, which is an overestimate.

4. The radius of a spherical balloon is increasing at a constant rate of 2 cm/sec. At the instant when the radius is 3 cm, what is the rate of change of the volume of the balloon, in cm^3/sec ? (The volume of a sphere is $V = \frac{4}{3}\pi r^3$).

- (A) 24π

(B) 36π

(C) 72π

(D) 144π

5. What is the value of $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x}{e^x + x^2}$?

(A) 0

(B) 3

(C) $\frac{3}{e}$

(D) ∞

6. The amount of water in a storage tank, in gallons, is modeled by a continuous function $W(t)$ for $0 \leq t \leq 12$, where t is measured in hours. It is given that $W'(5) = -3$. Which of the following is the best interpretation of this statement?

(A) During the first 5 hours, the amount of water in the tank decreased by 3 gallons.

(B) At time $t = 5$ hours, the amount of water in the tank is decreasing at a rate of 3 gallons per hour.

(C) At time $t = 5$ hours, the amount of water in the tank is 3 gallons.

(D) Between $t = 2$ and $t = 5$ hours, the amount of water in the tank decreased at an average rate of 3 gallons per hour.

7. A particle moves along a straight line with velocity given by $v(t) = t^2 - 4t + 3$ for time $t \geq 0$. On which of the following intervals is the speed of the particle increasing?

(A) $1 < t < 3$

(B) $t > 2$

(C) $1 < t < 2$ and $t > 3$

(D) $0 < t < 1$ and $2 < t < 3$

8. Car A is traveling due north at 40 miles per hour and Car B is traveling due east at 30 miles per hour. Both cars leave from the same intersection at the same time. What is the rate of change of the distance between the two cars, in miles per hour?

(A) 10

(B) 50

(C) 70

(D) 1200

Section 2: Free Response Questions

Show all your work clearly.

FRQ 1. A particle moves along the x-axis so that its velocity $v(t)$ at time $t > 0$ is given by $v(t) = t^2 - 6t + 8$.

- (a) Find the acceleration $a(t)$ of the particle at time t .
- (b) At what time(s) does the particle change direction? Justify your answer.
- (c) At $t = 1$, is the speed of the particle increasing or decreasing? Give a reason for your answer.

FRQ 2. A 15-foot ladder is leaning against a vertical wall. The base of the ladder is pulled away from the wall at a constant rate of 3 ft/sec.

- (a) How fast is the top of the ladder sliding down the wall when the base of the ladder is 9 ft from the wall? Show the steps that lead to your answer.
- (b) Let θ be the angle formed between the ladder and the floor. Find the rate of change of θ with respect to time when the base of the ladder is 9 ft from the wall. (Include appropriate units in your answer).